INTRODUCTION TO DATA MANAGEMENT

PROJECT REPORT

(Project Semester January-April 2025)

Crash Report Dashboard: Analysis of Accidents in US (2015-2024)

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Registration No: 12301024

Programme and Section: Btech-CSE, KM005

Course Code: INT 217

Under the Guidance of

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UID: 27952

Discipline of CSE/IT

Lovely School of Computer Science

Lovely Professional University, Phagwara

<u>CERTIFICATE</u>

This is to certify that **Soam Parkash** bearing **Registration no. 12301024** has completed INT 217 project titled, **“Crash Report Dashboard: Analysis of Accidents in US (2015-2024)”** under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of …………………………………………….

Lovely Professional University

Phagwara, Punjab.

Date:

DECLARATION

I, **Soam Parkash**, student of Introduction To Data Management under CSE/IT Discipline at, **Lovely Professional University, Punjab**, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12/04/2025 Soam Parkash

Registration No. 12301024 **Signature**

1. Introduction

This report provides a comprehensive analysis of traffic incidents documented in the dataset titled Crash_Reporting_-_Drivers_Data.xlsx. The data encompasses various crash reports recorded by multiple agencies, including Montgomery County Police, Takoma Park Police Department, and

Gaithersburg Police Department. The dataset spans incidents from 2015 to 2024 and includes detailed information on crash types, weather conditions, road characteristics, vehicle details, and driver behavior.

The primary focus of this report is to examine trends and patterns in collision types, injury severity, environmental factors, and vehicle involvement. Key attributes such as crash dates, road names, collision types, weather conditions, light conditions, traffic controls, and driver-related factors will be analyzed to identify recurring issues and potential areas for improvement in road safety. Additionally, the report will highlight the extent of vehicle damage and movement patterns during crashes.

2. Source of Dataset

The data we used comes from a file called **"Crash_Reporting_-_Drivers_Data.xlsx"**, which is available on a website called data.gov. It includes reports from the **Montgomery County Police**, as well as the **Gaithersburg, Rockville**, and **Takoma Park Police Departments**. There are **14,744 crash reports** in the file, with details like the date of the crash, where it happened, the kind of crash, how bad the injury was, the weather, and other important information.

3. Dataset Preprocessing

- **Rows and Columns**: The dataset contains **14,744 entries** and **37 columns**.
- **Missing Data**: Some values like road names or weather were missing or written badly (like "N/A" or in all caps).
- **Data Types**: Includes integers, datetime, and categorical (string) values.
- **Date Formatting**: Dates are standardized in the Crash Date column.

- **Cleaning Steps**:
    - Converted appropriate fields to date and category formats.
    - Handled missing values for analysis using imputation or exclusion, depending on context.
    - Trimmed and standardized categorical entries.

4. Analysis on Dataset (for each objective)

Analysis 1: Crash Frequency Over Time

i. General Description

We checked how many crashes happen each month to see if there's a pattern or specific time when crashes are more common.

ii. Specific Requirements

- Grouped the data by month using the cleaned crash date.
- Looked at months with more or fewer crashes.
- Made sure dates were correct.

iii. Analysis Results

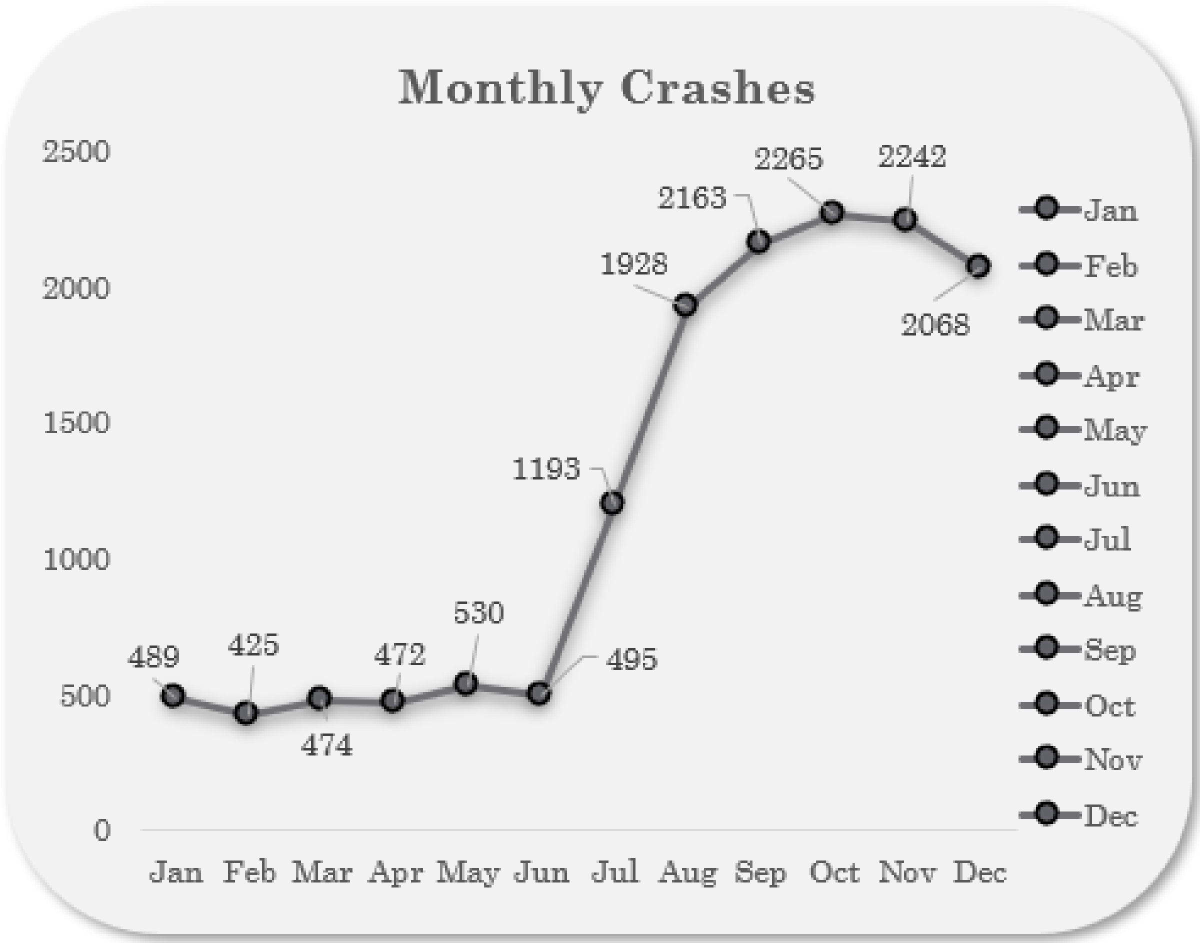
Here’s how many crashes happened each month:

- January: 489
- February: 425
- March: 474
- April: 472
- May: 530
- June: 495
- July: 1,193
- August: 1,928
- September: 2,163
- October: 2,265
- November: 2,242
- December: 2,068

We noticed more crashes happen from **July to December**, especially in **October and November**. **February** had the least crashes.

iv. Visualization

We made a **line chart** showing months on the X-axis and number of crashes on the Y-axis. The line goes up from January and reaches the top in October.



Analysis 2: Geographic and Agency-Specific Insights

i. General Description

We looked at where crashes happen the most and which police department reports the most crashes.

ii. Specific Requirements

- Used road names, towns, and coordinates.
- Cleaned location names so they're all the same.
- Focused on roads with at least 100 crashes.

iii. Analysis Results

Top 5 Roads with Most Crashes:

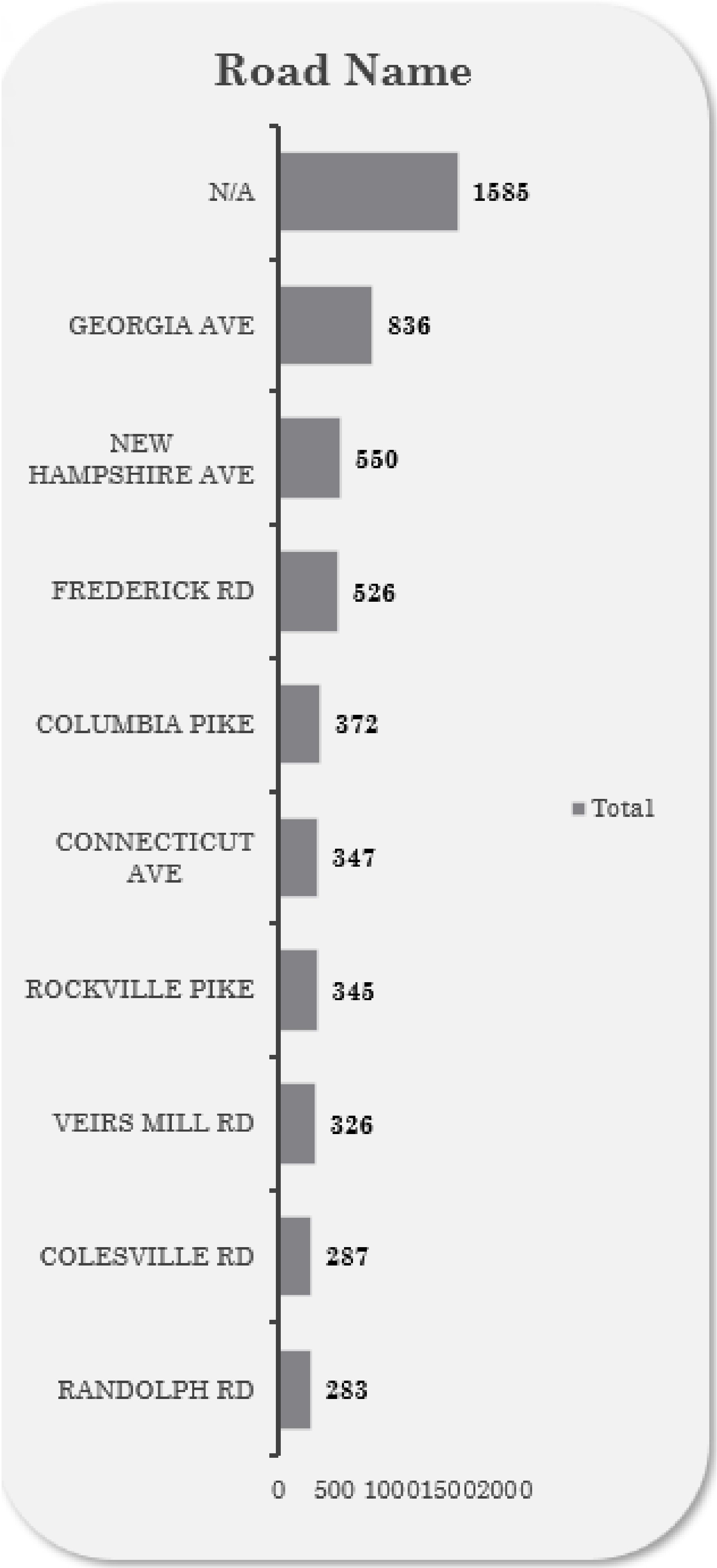
1. Georgia Ave: 836
2. New Hampshire Ave: 550
3. Frederick Rd: 526
4. Columbia Pike: 372
5. Connecticut Ave: 347

Reporting Agencies:

- Montgomery County Police: 12,246 crashes
- Gaithersburg Police: 858
- Rockville Police: 805
- Takoma Park Police: 301

iv. Visualization

We made a **bar chart** with road names and number of crashes. Georgia Ave had the biggest bar.


Road Name
N/A
1585
GEORGIA AVE
836
NEW HAMPSHIRE AVE
550
FREDERICK RD
526
COLUMBIA PIKE
372
CONNECTICUT AVE
347
ROCKVILLE PIKE
345
VEIRS MILL RD
326
COLESVILLE RD
287
RANDOLPH RD
283
Total
0
500
1000
1500
2000

Analysis 3: Injury Severity Monitoring

i. General Description

We checked how many crashes caused injuries and how bad the injuries were.

ii. Specific Requirements

- Cleaned the injury severity categories.
- Found out how many crashes had each type of injury.

iii. Analysis Results

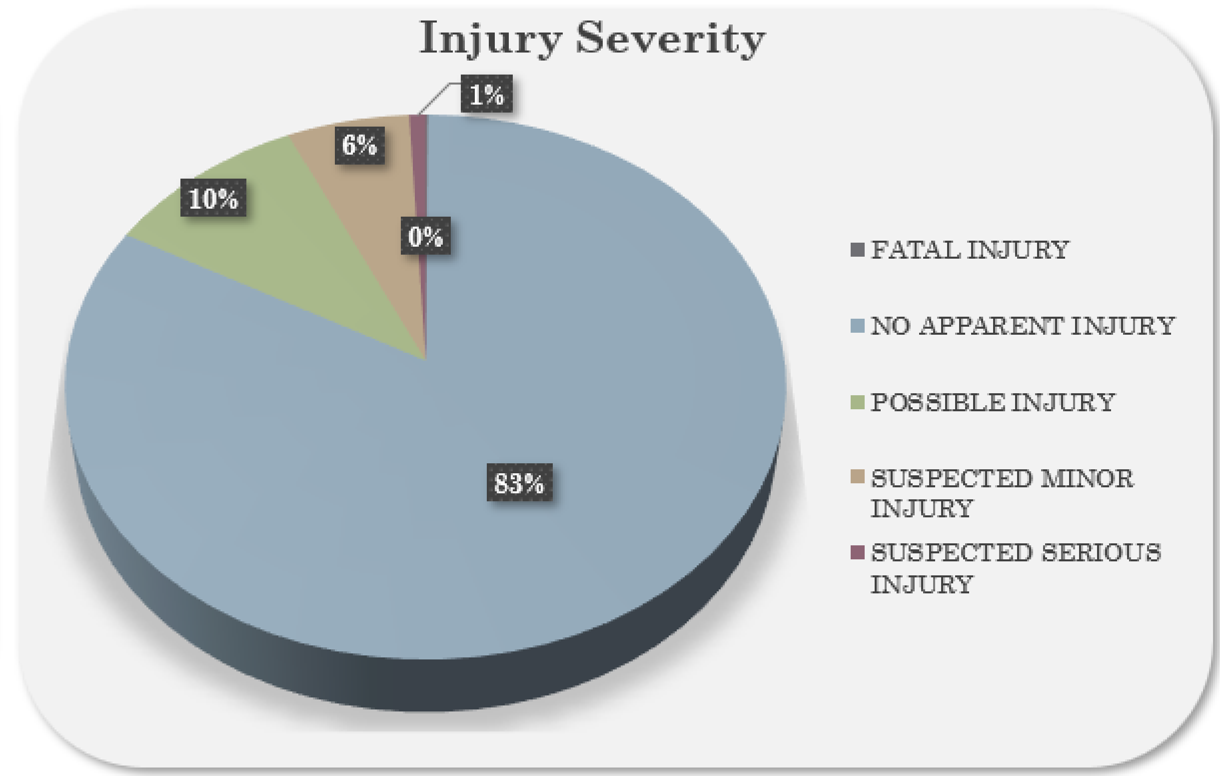
Injury Severity Breakdown:

- No Injury: 12,234 (83%)
- Possible Injury: 1,481 (10.1%)
- Minor Injury: 888 (6%)
- Serious Injury: 119 (0.8%)
- Fatal Injury: 14 (0.1%)

Most crashes had **no injury**, but some caused pain or serious harm.

iv. Visualization

We made a **pie chart**. The biggest part was "No Injury," and small slices showed serious and fatal injuries.



Analysis 4: Speed Limit Compliance Analysis

i. General Description

We checked the speed limits where crashes happened the most.

ii. Specific Requirements

- Checked speed limit values and removed any wrong ones.
- Grouped crashes by different speed limits.

iii. Analysis Results

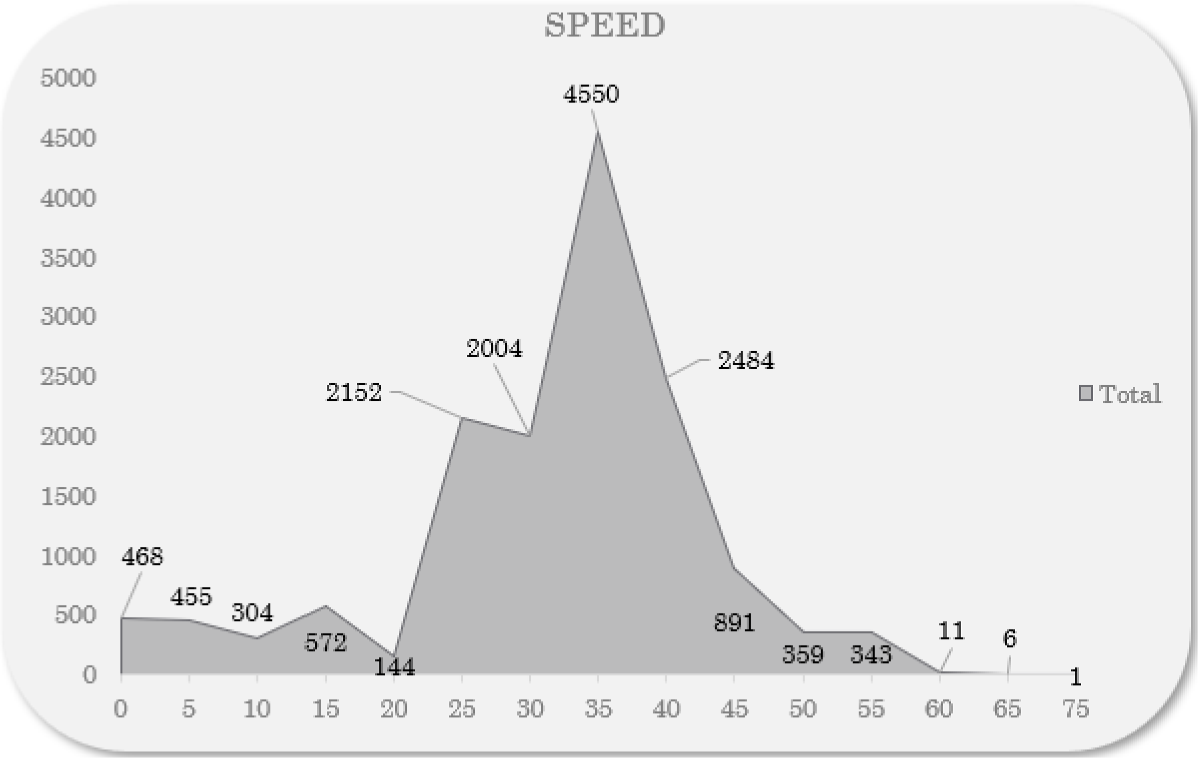
Crashes by Speed Limit:

- 0 mph: 468
- 5–15 mph: 1,331
- 25 mph: 2,152
- 30 mph: 2,004
- 35 mph: 4,550 (most)
- 40 mph: 2,484
- 45 mph: 891
- 50–75 mph: 719

Most crashes happened in **35 mph** zones.

iv. Visualization

We used an **area chart**. The graph bulged at 35 mph, showing that's where crashes happened the most.



Analysis 5: Crash Analysis Based on Light Condition

i. General Description

We checked if lighting conditions (like daylight or dark) made a difference in crashes.

ii. Specific Requirements

- Used categories like Daylight, Dark with Lights, Dark without Lights, Dawn, Dusk.
- Cleaned up entries like “Unknown.”

iii. Analysis Results

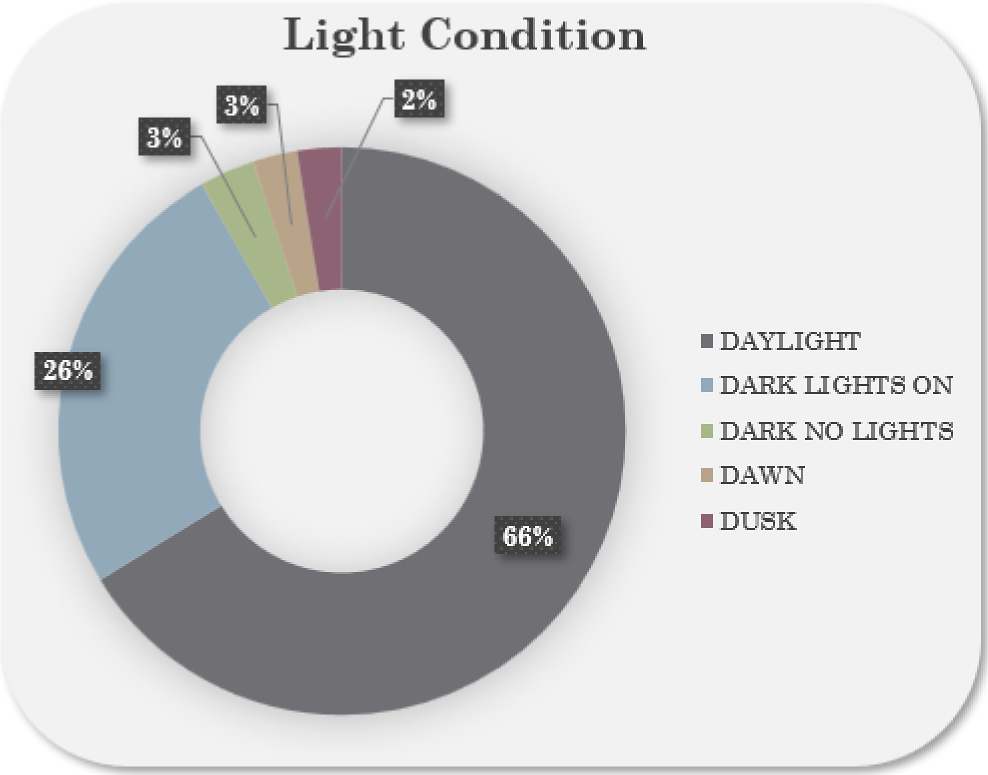
Crashes by Light Condition:

- Daylight: 9,760 (66.2%)
- Dark with Lights: 3,772
- Dark without Lights: 463
- Dawn: 373
- Dusk: 372

Most crashes happened during the **day**, but **dark roads without lights** had a higher chance of serious crashes.

iv. Visualization

We made a **column chart**. Daylight had the tallest bar, and dark no lights had a small but important bar.



5. Conclusion

After studying the crash data in Montgomery County, we learned many important things. Crashes are not spread out evenly during the year—they happen more often from **July to December**, especially in **October and November**. This tells us that during these months, drivers and authorities should be extra careful.

Some roads like **Georgia Ave, New Hampshire Ave, and Frederick Rd** had way more crashes than others. These places might need better road signs, traffic signals, or maybe even speed bumps to stop accidents.

Most of the crashes didn’t cause injuries, which is good. But still, there were a few crashes that caused serious or even fatal injuries, so we must keep working to make roads safer.

We also found that **35 mph** speed zones had the highest number of crashes. This might mean that even in medium-speed areas, people are driving too fast or not paying enough attention.

Lastly, we saw that **most crashes happened in daylight**, probably because more cars are on the road then. But crashes that happened in **dark areas without lights** were more dangerous. So, adding streetlights in dark spots could help a lot.

6. Future Scope

Even though our analysis gave us many useful ideas, there’s still a lot more we can do in the future.

One cool idea is to **add real-time data** from traffic cameras or police updates. This could help predict when and where crashes might happen, and police can reach the spot faster to help people.

We can also use **machine learning**, where computers learn from old crash data and help predict dangerous roads or times of the day. This would make the system smarter and more helpful over time.

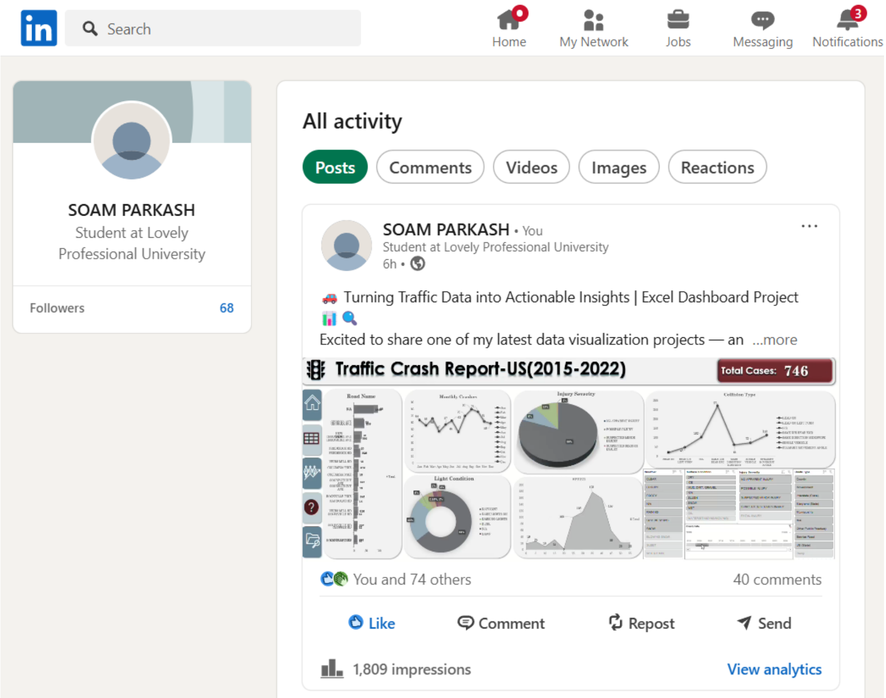
If we include extra information—like **weather**, **driver behaviour** (like texting or drinking), **types of vehicles**, and **how good or bad the roads are**—then we can understand crashes even better and find new ways to prevent them.

So, this report is just the beginning. With better tools and more data, we can make roads safer for everyone!

7. References

- **File Used**: Crash_Reporting_-_Drivers_Data.xlsx
- **Source**: “Crash Reporting Drivers Dataset,”
- **Tool Used**: Microsoft Excel (for cleaning, analysis, and charts)

❖ LinkedIn



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Traffic Crash Report-US(2015-2022)
Total Cases: 746
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